

right; and pulling backwards activated the brakes. Activating a switch on the handle before pulling back would put the LRV into reverse. Pulling the handle all the way back activated a handbrake.

The control and display modules were situated in front of the handle and gave information on the speed, heading, pitch, and power and temperature levels. Navigation was based on continuously recording direction and distance through use of a directional gyro and odometer, and inputting this data to a computer which would keep track of the overall direction and distance back to the LM. There was also a Sun-shadow device which could give a manual heading based on the direction of the Sun, using the fact that the Sun moved very slowly in the sky.

The LRV was deployed using a system of pulleys and braked reels using ropes and cloth tapes. The rover was folded and stored in quad 1 with the underside of the chassis facing out. One astronaut would climb the egress ladder on the Lunar Module and release the rover, which would then be slowly tilted out by the second astronaut on the ground through the use of reels and tapes. As the rover was let down from the bay, most of the deployment was automatic. The rear wheels folded out and locked in place and when they touched the ground the front of the rover could be unfolded, the wheels deployed, and the entire frame let down to the surface by pulleys. The rover components locked into place upon opening. Cabling, pins, and tripods would then be removed and the seats and footrests would be raised. After switching on all the electronics, the vehicle was ready to back away from the Lunar Module.

A total of four lunar rovers were built, one each for Apollos 15, 16 and 17, and one that was used for spare parts after the cancellation of further Apollo missions. The LRV was developed in only 17 months and yet performed all its functions on the Moon with no major anomalies.

Harrison Schmitt of Apollo 17 said, “The Lunar Rover proved to be the reliable, safe and flexible lunar exploration vehicle we expected it to be. Without it, the major scientific discoveries of Apollo 15, 16, and 17 would not have been possible; and our current understanding of lunar evolution would not have been possible.” 